

Assignment-1 Solution

1. Redundancy means including extra hardware components while designing a system so that in case of a failure, the system can still operate normally. It increases the reliability of the system.

Example: For data storage, RAID incorporates redundancy by storing data across multiple disks, so if one disk fails, the system can still access data from the other disks.

Any other relevant example

2.
 - a. We use the geometric mean because it avoids results being skewed by very high or very low benchmark scores. It gives a fairer picture of the system's overall performance than a simple average.
 - b. A SPEC ratio greater than 1 means the tested system performs **better or faster** than the reference system.

c.

$$SPEC\ Ratio = \sqrt[4]{\frac{250}{200} * \frac{180}{120} * \frac{300}{400} * \frac{90}{60}} = 1.205$$

CPU Z is around 20% faster than the reference system.

3.
 - a. CPU Time (Alpha) = IC x CPI x Clock Cycle Time
= $1.5 \times 10^9 \times 3.2 \times 2 \times 10^{-9}$
= 9.6 seconds
CPU Time (Beta) = IC x CPI x Clock Cycle Time
= $1.5 \times 10^9 \times 2.8 \times 2.5 \times 10^{-9}$
= 10.5 seconds
 - b. Alpha CPU is faster by 0.9 seconds (10.5-9.6).
 - c. Alpha CPU would be ideal, since a lower CPU time means lower latency and faster responsiveness.

4.
 - a. Proper explanation using own words. Check the slides and text textbook for reference.
 - b. Make the common case faster: Amdahl's Law focuses optimization efforts on the parts of the program where most time is spent (common case), because speeding up rarely executed parts has little impact.
 - c. Color correction time = 40% x 10s = 4s
 $T(\text{improved}) = T(\text{affected})/n + T(\text{unaffected})$
= $4/4 + 6$
= 7 seconds
It is 3 seconds/30% faster compared to the initial time.

5.

a. Avg. CPI (CPU X) = $0.4 \cdot 3 + 0.4 \cdot 4 + 0.2 \cdot 2 = 3.2$
Avg. CPI (CPU Y) = $0.4 \cdot 2 + 0.4 \cdot 3 + 0.2 \cdot 2 = 2.4$

b. Total Clock Cycles = Instruction count x Average CPI
CPU X clock cycles = $5 \cdot 10^5 \cdot 3.2 = 1600000$
CPU Y clock cycles = $5 \cdot 10^5 \cdot 2.4 = 1200000$